

Table 4: Characteristics of bitmap-indexed attributes on IMDb and legacy-comparable backend-native compressed-payload sizes (MiB; entries below 0.001 MiB are shown in bytes). These sizes are not complete serialized-file or in-memory-object footprints. DDC uses the fixed-zero $S = 4,096$ core-payload protocol; the baseline columns use their native payload counters. CRoaring reports its run-optimized native payload after `runOptimize()`. N is the source relation’s row count. Clustering (CF_1) is the pooled average length of maximal consecutive 1-runs. Density is the average bitmap density $M/(NC)$. Frequency Gini G is computed from all observed category counts using the same definition as in R2.W1. Inc. is the relative size change from DDC. Negative values indicate a smaller baseline.

Bitmap	N	Card.	Density	CF_1	Gini	DDC	WAH		EWAH		CRoaring (runOptimize)	
						Size	Size	Inc.	Size	Inc.	Size	Inc.
<i>movie_companies.company_type_id</i>	2.6M	2	0.5	1.3M	0.012	0.3	40 B	$\approx -100\%$	56 B	$\approx -100\%$	590 B	$\approx -100\%$
<i>title.kind_id</i>	2.5M	6	0.2	4	0.619	0.6	0.8	+33%	0.8	+33%	0.8	+25%
<i>cast_info.role_id</i>	36.2M	11	0.09	2.8M	0.547	4.3	308 B	$\approx -100\%$	432 B	$\approx -100\%$	0.008	$\approx -100\%$
<i>person_info.info_type_id</i>	3.0M	22	0.05	2	0.543	2.3	4.7	+1.0×	5.6	+1.4×	3.5	+56%
<i>movie_info.info_type_id</i>	14.8M	71	0.01	35	0.837	2.2	1.3	−41%	1.4	−36%	0.9	−60%
<i>title.production_year</i>	2.5M	132	0.007	1	0.696	3.7	10.0	+1.7×	14.0	+2.8×	4.7	+26%
<i>company_name.country_code</i>	235.0K	215	0.004	1	0.919	0.2	0.6	+2.0×	0.8	+3.0×	0.3	+20%
<i>movie_keyword.movie_id</i>	4.5M	476K	2×10^{-6}	9	0.688	2.4	9.5	+3.0×	11.4	+3.8×	6.3	+1.6×
<i>cast_info.movie_id</i>	36.2M	2.3M	4×10^{-7}	1	0.529	134.7	296.9	+1.2×	557.5	+3.1×	149.3	+11%
<i>cast_info.person_id</i>	36.2M	4.1M	2×10^{-7}	7	0.805	24.2	88.7	+2.7×	113.6	+3.7×	52.9	+1.2×
Total	—	—	—	—	—	174.9	412.4	+1.4×	705.1	+3.0×	218.6	+25%